An Observation Plane for the Riemann Hypothesis: Judgement Completeness and the Open Reading Constraint

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Abstract

The Riemann hypothesis asserts that all non-trivial zeros of ζ lie on the critical line $\Re s = 1/2$. This paper proves nothing; it does two things: it restates this assertion on a newly constructed observation plane, and it states, item by item, which parts of the restatement are machine-verified and which remain open. The construction has two steps. The first is a base-point migration: the operator $T(z) = 1/(z-i)-1$ turns the critical line into the unit circle $|T| = 1$, and gives the exact mark $i^{1/(2i)} = e^{\pi/4}$ of this line in the new coordinates. The second step rotates the coordinate axes by 45°, producing fold coordinates (u, v) equivalent to the standard complex coordinates. On this observation plane, three coordinate properties of the projection $d = \sqrt{2}(\Re s - \frac{1}{2})$ are verified: the critical line is mapped to a single point; no point off the critical line is mapped to that point; and the definition of the projection does not involve ζ. On top of this, judgement completeness is verified: for every zero in the strip, "on the critical line" and "projection reading zero" are equivalent, zero-ness itself is characterized by the two projection components of the value, and the position along the line is never used. From these equivalences, the Riemann hypothesis is equivalent to a universal statement: the projection reading of every zero in the strip is zero. The equivalences are verified in Lean 4; the universal statement remains open. Every claim in the paper compiles under Lean 4 and mathlib (0 error, 0 `sorry`), and no axioms beyond those of mathlib are introduced; classical results that are not machine-verified are marked KNOWN and given references. The paper contains four figures.

1 Introduction: the reduction chain and its boundary

Phenomenon. The usual form of the Riemann hypothesis is: all non-trivial zeros of ζ lie on the critical line $\Re s = 1/2$ [1, 10, 11]. Around this statement, the classical theory has already produced several families of mutually equivalent restatements, for example the reflection pairing of zeros about the critical line, the absence of zeros outside the critical strip, and the contrapositive form "the hypothesis fails if and only if an off-line zero exists" [9]. The companion assembly work has written this reduction chain step by step as Lean 4 declarations. The last step of the chain is "an off-line zero yields a contradiction"; for the known families of bridges (Euler product, functional equation, phase, growth), this step does not go through: an off-line zero drags along the four zeros of its reflection orbit, and bridges whose input condition is $\zeta \neq 0$ cannot be used anywhere on that orbit (the verification is given in §8).

Theorem/lemma (cited). Two verified equivalence items (both machine-checked versions of standard facts) reduce the scope of the problem to "whether an off-line zero exists inside the critical strip":

- `rh_iff_all_zeros_on_line_in_strips` (proof.lean): the Riemann hypothesis holds if and only if for every $T \geq 0$, every zero in the strip ($0 \leq \Re \leq 1$, $|\Im| \leq T$) satisfies $\Re = 1/2$;
- `rh_failure_iff_offline_common_zero` (zeta_rh_oddeven_iso.lean): the hypothesis fails if and only if there exists an off-line zero in the strip ($0 < \Re < 1$, $\Re \neq 1/2$, $\Im \neq 0$).

Hypotheses. Absence of zeros outside the strip and the symmetry of the functional equation are classical; we cite verified versions. Every classical result that is not machine-verified is marked KNOWN at its point of use and given a reference.

Lean formalization. The reduction items cited in this section are distributed over `proof.lean`, `ValueBridge.lean` and `zeta_rh_oddeven_iso.lean`, all with 0 `sorry`. The counting equivalence (the Riemann hypothesis holds if and only if the number of zeros on the line equals the number of zeros in the strip, `rh_iff_N0_eq_N`, ValueBridge.lean) reappears in §7.

Conclusion. The task can thus be stated in one concrete sentence: build an observation surface on which "whether a zero lies on the critical line" is fully decided, and on which "all zeros lie on the line" becomes a universal statement about this surface. This paper does this, and writes down the boundary as well. Each section of the paper gives five parts in order: phenomenon, theorem/lemma, hypotheses, Lean formalization, conclusion.

The contributions of this paper are listed in three layers.

- Mathematical layer (organization of known material): the observation plane and the fold coordinates (a linear change of basis of the standard coordinates, §2 to §4); the reading-form equivalence of the Riemann hypothesis, together with the counting and common-vertical-line equivalences (§7).
- Formalization layer (the machine-verified results of this draft): the Lean 4 verification of the constructions and of the equivalence chain above (§9); the precise form of judgement completeness: for every zero in the strip, being on the line and the reading vanishing are true together and false together (§5 and §6).
- Methodological layer (diagnosis and calibration of this draft): the two-level calibration of judgement (the individual judgement is complete, the universal judgement is open, and there is no independent intermediate level; remark at the end of §7); the structural diagnosis and the boundary of the remaining step (§8).

2 Base-point migration: the critical line is the unit circle

Phenomenon. In standard coordinates the critical line is the straight line $\Re s = 1/2$, unrelated to the origin of the reals; the functional-equation reflection $s \mapsto 1-\bar{s}$ has its fixed points on this line. The construction takes its base points from the equidistance structure of this line: the critical line is the perpendicular bisector of the base points i and $1+i$ (`critical_line_equidistant_basepoint_i`), and also the perpendicular bisector of the origin 0 and the unit point 1 (`on_line_iff_equidistant_base_one`). To move the observation position onto this line, take the operator

$$T(z) = \frac{1}{z-i} - 1$$

(this is a Möbius transformation: a reciprocal and a translation composed; first reciprocal about the centre i, then translate by -1). The modulus of T is the ratio of the distances to the two base points $|T(z)|^2 = |z-1-i|^2/|z-i|^2$ (`T_normSq_ratio`), and equals 1 on the equidistance line: the critical line becomes the unit circle $|T| = 1$ in the new coordinates; reflection corresponds to inversion in the unit circle; the real number 0 is no longer a special point ($T(0) = -1+i$, outside the circle). The exact position of this line in the new coordinates is the mark $i^{1/(2i)} = e^{\pi/4}$.

Theorem/lemma. Three verified results:

Theorem 2.1 (Base-point migration)**.** *For $z \neq i$: $|T(z)|^2 = 1$ if and only if $\Re z = 1/2$ (`T_normSq_eq_one_iff`); for $z \neq i$, $1+i$: $T(1-\bar{z}) = 1/\overline{T(z)}$, i.e. reflection is inversion in the unit circle (`T_reflect`); and the mark formula $i^{1/(2i)} = e^{\pi/4}$, the half-mark $\frac{1}{2}i^{1/(2i)} = e^{\pi/4}/2$, dual inverses $e^{\pi/4}e^{-\pi/4} = 1$ (`i_pow_half_inv_two_i`, `half_succ_position`, `pair_inverse`).*

Hypotheses. T is an invertible coordinate change (the inverse operator `TInv` is verified as well); the computation is exact, with no numerical approximation; the complex power takes the principal branch, and this choice is made explicit in the proof (argument cancellation $\frac{1}{2i} \cdot \frac{i\pi}{2} = \frac{\pi}{4}$). This section and the following ones do not take i or $\pi/2$-type transformations as construction input; they appear only in the human-readable read-back layer.

Lean formalization. `PreciseMigration.lean`: `T`, `TInv`, `T_normSq_eq_one_iff`, `T_reflect`, `i_pow_i`, `i_pow_half_inv_two_i`, `half_succ_position`, `pair_inverse`; 20 declarations in total, 0 `sorry`.

Conclusion. The critical line acquires two readings: "the unit circle" and "the mark $e^{\pi/4}$". Judging $\Re = 1/2$ is from now on a judgement on a circle. Figure 1 shows the two circles attached to the base points; the second step of the construction is in the next section (§3): rotate the axes by 45° to obtain fold coordinates.

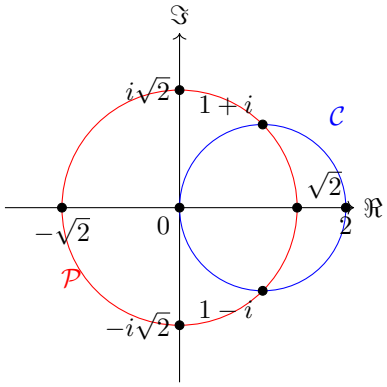


Figure 1: The two base-point circles (drawn for $p = 2$): the critical circle $\mathcal{C} = \{w : \|w - 1\| = 1\}$ (centre 1, radius 1, passing through 0 and 2) and the prime circle $\mathcal{P} = \{z : \|z\|^2 = p\}$ (centre 0, radius $\sqrt{p}$; where the two-squares lattice points of the prime p lie); the two circles meet at $1 \pm i$.

3 Fold coordinates: four readings

Phenomenon. Rotating the axes of the standard complex plane by 45° gives two orthogonal coordinate axes u and v (constant $\sqrt{2}/2 = 2^{-1/2}$). In these coordinates the imaginary axis ($\Re z = 0$) presents itself as a 90° fold: two straight arms run out of -1 at $\pm 45°$, passing through $\pm i$ (drawn up to $\pm i$); the fold (the union of the two rays) is homeomorphic to the real axis. One and the same unit circle has one appearance per reading: the standard complex plane, the $\pm 45°$ rotated frame, the presentation reading in which each axis carries only the unit 1, and the fold reading (Figure 2).

Theorem/lemma. Three verified results:

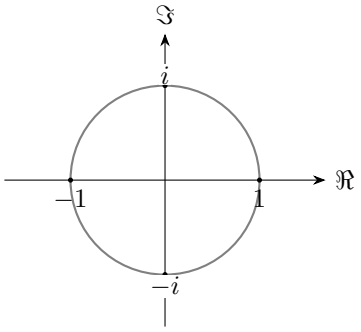
Theorem 3.1 (Fold coordinates)**.** *(i) Equivalence with the standard coordinates:* $\frac{u+v}{\sqrt{2}} = \Re z$, $\frac{u-v}{\sqrt{2}} = \Im z$ *(`axis_recovery_re`, `axis_recovery_im`); the coordinates determine the position uniquely (`axis_unique`); modulus preservation* $u^2 + v^2 = (\Re z)^2 + (\Im z)^2$ *(`axis_norm_sq_preserving`); conversely every pair* (u, v) *corresponds to some complex position (`axis_exists`, simply putting the standard coordinates back into the container). (ii) The imaginary axis is the diagonal:* $\Re z = 0$ *if and only if* $u + v = 0$ *(`fold_imag_axis_diag`); the fold meets the imaginary axis at* $\pm i$ *(`fold_vertices_are_imag_units`); the fold is homeomorphic to the real axis (`foldArcHomeo`). (iii) Triangle reading of the fold: the two legs each have length* $\sqrt{2}/2$*; the hypotenuse corresponds to the length reading* $|i| = 1$ *and the value reading* $[0, -1]$ *(value* -1*); Pythagorean relation* $(\sqrt{2}/2)^2 \cdot 2 = 1^2$ *(`fold_tri_pythagorean`); arm sum* $0.5i + 0.5i = i$*, value chain* $i \cdot i = -1$ *(`fold_tri_arms_sum` etc.).*

Hypotheses. The fold is constructed only from standard facts (Euler's identity $e^{i\pi} = -1$ gives the negative-side starting point, rotation gives $\pm i$, direction 45°, length $\sqrt{2}$); it contains no new mathematics; it is a restatement of the imaginary axis (the fold meets the imaginary axis at $\pm i$, `fold_vertices_are_imag_units`; the line generated by the two intersection points is the imaginary axis, `imag_axis_eq_span_fold_vertices`). The fold and "the native straight axis in a raised dimension" are two options at the

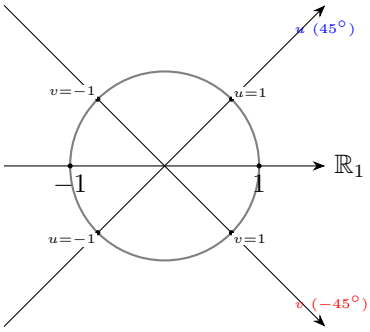
presentation layer: the straight axis raises the dimension by one more level and produces no new machine-checkable content; the fold keeps the dimension and makes explicit, and machine-checkable, the fact that in the standard framework the two roots enter the plane differently ($\sqrt{2}$ stays on the real axis, $\sqrt{-1}$ enters by raising the dimension). This paper takes the fold; this is a choice at the presentation layer, stated openly for the reader to judge. The fold reading comes from the author's published representation framework [15, 16]; only the coordinate part of it is used here.

Lean formalization. `EulerFoldImagAxis.lean` (89 declarations) and `NativeAxisEquiv.lean` (11 declarations) cover the items above; all 0 `sorry`, no axioms introduced.

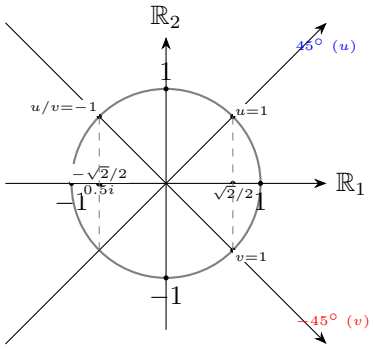
Conclusion. The complex plane acquires two mutually convertible coordinate systems. Four correspondences follow (all machine-checked): conjugation corresponds to swapping u and v (`fold_conj_swap`); the imaginary axis corresponds to the diagonal $u + v = 0$ (`fold_imag_axis_diag`); the critical line corresponds to the horizontal line $u + v = \sqrt{2}/2 = 2^{-1/2}$ (`fold_critical_line_horiz`); real values correspond to the fixed line of the swap (`fold_real_iff_swap_fixed`). The next section takes the observation plane in these two coordinate systems (§4).



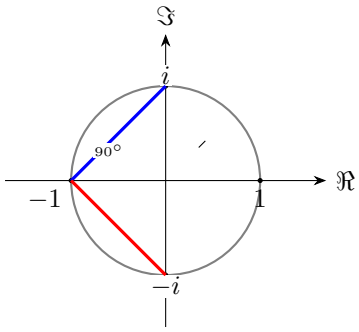
(a) complex-plane view: standard $\Re/\Im$ axes, ± 1, $\pm i$, unit circle



(b) the same circle in the $\pm 45°$ rotated frame: the u/v axes and their unit points $u = \pm 1$, $v = \pm 1$



(c) presentation reading: the u/v axes (unit 1, no numbers on the axes); the readings on the real axis $\sqrt{2}/2$ (the $+1$ reading) and $0.5i$ (-1 reading, the vertical component of the arm), with feet at $\pm\sqrt{2}/2$



(d) the $90°$ fold starting at -1: two straight arms through $\pm i$ (drawn up to $\pm i$)

Figure 2: One unit circle under four readings: (a) complex-plane view (standard $\Re/\Im$ axes, ± 1, $\pm i$); (b) the same circle in the $\pm 45°$ rotated frame — the u/v axes and their unit points $u = \pm 1$, $v = \pm 1$; (c) presentation reading: the u/v axes carry only the unit 1, while every value appears on the real axis through perpendicular projection — the readings are $\sqrt{2}/2$ (the $+1$ reading) and $0.5i$ (the -1 reading, the vertical component of the arm), with feet at $\pm\sqrt{2}/2$; (d) the $90°$ fold starting at -1: two straight arms through $\pm i$ (drawn up to $\pm i$). All radii are 1; schematic.

4 The observation plane: three coordinate properties

Phenomenon. For a single point, "is it on the critical line" is answered by one number: the distance to the critical line $\Re s - 1/2$. The position along the line $\Im s$ takes no part in this judgement. The observation plane takes two axes: axis 1 along the critical line, reading $\sqrt{2}\,\Im s$; axis 2 perpendicular to the critical line, reading $\sqrt{2}(\Re s - \frac{1}{2})$, whose origin is placed on the critical line. The projection is the reading of axis 2, written

$$d(s) = \texttt{foldObsProj}(s) = u(s) + v(s) - \frac{\sqrt{2}}{2} = \sqrt{2}\Big(\Re s - \frac{1}{2}\Big).$$

Theorem/lemma. Three coordinate properties are verified:

Theorem 4.1 (Projection)**.** *(1) The projection collapses to a point: if $\Re s = 1/2$ then $d(s) = 0$, i.e. the whole critical line is mapped to a single point on axis 2 (`critical_projects_to_axis2_point`; fold-coordinate version `fold_critical_projects_point`). (2) Faithfulness: conversely, $d(s) = 0$ if and only if $\Re s = 1/2$, for all $s \in \mathbb{C}$, not only in the strip (`only_critical_projects_to_point`; `fold_projection_point_iff_critical`). (3) No self-reference: the definition of d uses only the fold coordinates, not ζ (`foldObsProj_independent_of_zeta`; standard version `obsProj_independent_of_zeta`).*

Hypotheses. Properties (1)–(3) hold for every point, zero or not. They are coordinate facts and carry no information about zeros; the strip condition $0 < \Re s < 1$ is used only in §6, when zeros are judged.

Lean formalization. `AxisExactObservation.lean` (18 declarations) and `FoldObservation.lean` contain the items above; 0 `sorry`.

Conclusion. The projection is a qualified decision tool: the reading needed for the judgement exists, and its definition is independent of ζ. The judgement also needs a reading on the side of zero-ness, given in the next section (§5).

5 The projection characterization of zero-ness: two value-range anchors

Phenomenon. The judgement needs to answer another question: given a point s, how do we know that it is a zero of ζ. Applying the fold formula to the value $\zeta(s)$ produces two real components

$$U(s) = \frac{\sqrt{2}}{2}\big(\Re\zeta(s) + \Im\zeta(s)\big), \qquad V(s) = \frac{\sqrt{2}}{2}\big(\Re\zeta(s) - \Im\zeta(s)\big).$$

At a zero both components vanish simultaneously.

Theorem/lemma. Three verified results:

Theorem 5.1 (Value-range anchors)**.** *(1) Value anchors: $U + V = \sqrt{2}\,\Re\zeta(s)$, $U - V = \sqrt{2}\,\Im\zeta(s)$ (`zeta_fold_sum_re`, `zeta_fold_diff_im`). (2) The components are invertible: (U, V) recovers $\zeta(s)$ (`zeta_fold_invertible`), and $\zeta(s) = 0$ if and only if $U(s) = V(s) = 0$ (`zeta_fold_zero_iff`). (3) A zero forces both anchors to vanish: if $\zeta(\rho) = 0$ then $U(\rho) + V(\rho) = 0$ and $U(\rho) - V(\rho) = 0$ (`zeta_fold_dual_proj_zero_of_zero`).*

Hypotheses. These are pointwise identities; no extra hypotheses are needed; $\zeta(\rho) = 0$ is the premise when zero-ness is discussed.

Lean formalization. `FoldObservation.lean`: `zetaFoldU`, `zetaFoldV`, `zeta_fold_invertible`, `zeta_fold_zero_iff`, `zeta_fold_sum_re`, `zeta_fold_diff_im`, `zeta_fold_dual_proj_zero_of_zero`; 0 `sorry`.

Conclusion. Zero-ness is fully characterized by the two projection components of the value; combined with the coordinate reading of §4, this yields the judgement completeness of the next section (§6).

6 Judgement completeness: the weak assertion

Phenomenon. Now combine the two sides: zero-ness lives in the value components (U, V), being on the critical line lives in the axis-2 reading d, and the position along the line never appears. One may say: the information that remains after the projection drops axis 1 is enough to decide whether a zero lies on the critical line.

Theorem/lemma.

Theorem 6.1 (Judgement completeness; verified)**.** *For a zero ρ in the strip ($\zeta(\rho) = 0$, $0 < \Re\rho < 1$):*

$$\Re\rho = \tfrac{1}{2} \iff d(\rho) = 0 \iff \big(U(\rho) = 0 \wedge V(\rho) = 0 \wedge d(\rho) = 0\big).$$

(`projection_zero_line_complete`, FoldObservation.lean; the strip factor $2^{-\rho} \neq 1$ is verified separately; supporting item `fold_zero_judgement_preserved`.)

Hypotheses. The result of this section is usually called the weak assertion: it states the completeness of the judgement, not that the conclusion holds. "Complete" means that the judgement sentence and the projection form are true together and false together, derivable in both directions; it does not mean that "the reading equals zero" holds. Table 1 below separates the three kinds of statements: the first two rows are verified equivalences, the third row is the statement to be proved. The coordinate properties hold for every point, independent of zeros; judgement completeness holds for zeros, with $\zeta(\rho) = 0$ as the premise. Neither of them gives us the conclusion "all readings vanish": that requires the universal quantifier over all zeros (§7).

Statement	Content	Status
Coordinate properties	the critical line maps to a point; no other point maps there; definition free of ζ	verified (§4)
Judgement completeness	zeros in the strip: on the line iff $d = 0$; zero-ness characterized by the two value anchors	verified (§6)
Universal reading	every zero in the strip satisfies $d = 0$	open; equivalent to the Riemann hypothesis (§7)

Table 1: The three kinds of statements and their verification status.

Lean formalization. `FoldObservation.lean`: `projection_zero_line_complete` etc.; 0 `sorry`.

Conclusion. The work at the judgement layer is complete here: the criterion is lossless. What judgement each individual zero yields is the universal statement of the next section.

7 Equivalent restatements: the Riemann hypothesis as a reading constraint

Phenomenon. Substituting a zero into the judgement of §6 yields a universal statement: the projection reading of every zero in the strip is zero. This statement is true exactly when the Riemann hypothesis is.

Theorem/lemma.

Theorem 7.1 (Reading form; verified)**.** *The Riemann hypothesis (`rh_classical`) is equivalent to*

$$\forall s:\ 0 < \Re s < 1\ \wedge\ \Im s \neq 0\ \wedge\ \zeta(s) = 0\ \Longrightarrow\ \mathit{foldObsProj}(s) = 0.$$

(`fold_rh_iff_axis_form`, FoldObservation.lean; standard-coordinate version `rh_iff_axis_form`, AxisExactObservation.lean.)

Moreover several further equivalent formulations (equivalent restatements of the Riemann hypothesis have many precedents in the literature, e.g. [12]) are verified:

- Counting form: the Riemann hypothesis holds if and only if the number of zeros on the line equals the number of zeros in the strip (`rh_iff_N0_eq_N`; the version for each height T is `count_eq_iff_all_on_line`, ValueBridge.lean), whose geometric background is the argument-principle rectangle (Figure 3);
- Common-vertical-line form: the Riemann hypothesis holds if and only if all zeros in the strip lie on one single vertical line (`common_line_iff_RH`, probe CommonLineCheck.lean); here the "line" can only be $\Re = 1/2$, forced by the reflection pairing;
- Contrapositive form: the hypothesis fails if and only if an off-line zero in the strip exists (`rh_failure_iff_offline_common_zero`).

As a positioning, Table 2 places the nearest-neighbour forms side by side with the difference from this draft (the increment of this draft is in the two layers of machine verification and calibration).

Relation form	Source	Difference from this draft
Counting form: number of zeros on the line equals number of zeros in the strip	classical corollary of the argument principle (textbook [9])	this draft lists it alongside the reading form, and machine-verifies that the two are true together
Horizontal monotonicity ⇔ Riemann hypothesis	Sondow–Dumitrescu [17]; Matiyasevich–Saidak–Zvengrowski [18]	the form is a monotonicity of moduli; the form in this draft is the coordinate reading d
Elementary criterion (divisor function)	Lagarias [12]	the object is the divisor function, a different class from coordinate readings
Equivalence involving ζ'	Speiser [5]	the object is the derivative
Proportion of zeros on the line (1/3, 2/5)	Levinson [6]; Conrey [7]	partial results, not equivalences

Table 2: Positioning relative to the nearest neighbours (all known results).

Hypotheses. These equivalences are standard pointwise translations; no hypotheses are added. Watch the quantifier: the version truncated at height $|\Im| \leq T$ (saying "the zeros in the strip with height at most T are collinear") is a finite statement and does not imply the Riemann hypothesis; only the universal version is equivalent to it.

Lean formalization. `AxisExactObservation.lean`, `FoldObservation.lean`, `CommonLineCheck.lean` and `ValueBridge.lean` cover the items above; 0 `sorry`.

Conclusion. The Riemann hypothesis can be restated as one sentence about readings: the projection reading of every zero in the strip is zero. The restatement is verified; the sentence itself remains open; the structural diagnosis of the gap is in the next section (§8).

Remark 7.2 (Two-level calibration of judgement)**.** *Reading the verified results side by side gives the following facts. (i) Individual level: for each zero in the strip, "on the critical line" and "reading $d = 0$" are true together and false together, and the judgement does not refer to ζ (§6). (ii) Universal level: "all readings of zeros in the strip are zero" is true together with the Riemann hypothesis (§7). (iii) Intermediate: the common intermediate candidate "flip count equals winding count" ($K_{\text{flip}} = K_{\text{wind}}$) does not constitute an independent level — it is true together with the universal statement of (ii) (counting equivalence; its sufficient direction is verified by the conditional assembly, §8). Hence judgement in this framework has only two levels: the individual judgement is complete, the universal judgement is open, and there is no independent intermediate level. Also (standard observation, KNOWN, not relied upon in this paper): the negative direction of the universal statement has a finite certificate — a single off-line zero can be exhibited by an isolation region and a zero count (zero counting rests on the argument principle [9]); the positive direction has no purely instance-by-instance finite verification path, and must give an argument covering all cases.*

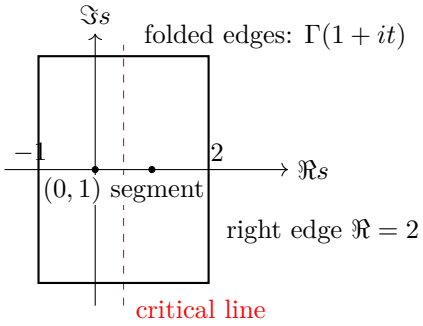


Figure 3: The argument-principle rectangle, with vertices $-1 \pm iT$, $2 \pm iT$ (drawn here for $T = 2$): the bottom edge $[-1, 2]$ (false-real-axis assembly: the negative half-plane is folded, via the functional equation, into the positive series region $\Re s > 1$), the right edge $\Re s = 2$ (bounded by $\sum n^{-2}$), and the top and left edges folded by the functional equation. The geometric background of the counting equivalence.

8 Structural diagnosis of the remaining step, and its boundary

Phenomenon. "No off-line zeros" cannot so far be derived from the existing structures; splitting the route "an off-line zero yields a contradiction" gives three verified structural diagnoses. (1) The bridge family fails on the off-line orbit: if ρ is an off-line zero in the strip, then ρ, $\bar{\rho}$, $1-\rho$, $1-\bar{\rho}$ are all zeros (reflection pairing, `zeta_zero_offline_pair`, proof.lean); bridges whose input condition is $\zeta \neq 0$ (Euler product, ratio, phase) are unusable anywhere on this orbit. At a zero the odd and even parts vanish simultaneously, and the single-lift ratio is $0/0$ (`self_reference_resolved`); the difference structure, which is the phase ratio, is well-defined but carries no information about ζ (`phase_ratio_unit`). (2) Linear observations cannot distinguish: the existing observables are linear or algebraic reorganizations of coordinates and preserve the zero set; under the assumption "an off-line zero exists", the eight-point signature (`eight_point_observation`) and the first-order skeleton (`offline_T_coord_skeleton`, TCoordOffline.lean) are all consistent with it and produce no contradiction. (3) The counting form is circular: writing the contradiction as a count, the only hypothesis left is "flip count equals winding count" ($K_{\text{flip}} = K_{\text{wind}}$), which is precisely the counting form of the Riemann hypothesis.

Theorem/lemma.

Theorem 8.1 (Conditional assembly; verified)**.** *Given the bridges on both sides of the count and the hypothesis $K_{\text{flip}} = K_{\text{wind}}$ (in Lean: the hypothesis `hflip_eq` of `value_bridge_assembly`), there are no off-line zeros (`value_bridge_assembly`, ValueBridge.lean; the contradiction is closed by `value_bridge_contradiction`). That is, the missing step of the chain is written as one counting identity; this identity is open, and it is equivalent to the Riemann hypothesis (the boundary of the reduction chain; see §7).*

Hypotheses. The input of the contradiction argument is "an off-line zero in the strip exists". Further, classical results for contrast (KNOWN): the zero-free region (de la Vallée Poussin) excludes zeros on the $\sigma \to 1$ side, but only up to a logarithmic distance [2, 3]; Hardy proved that there are infinitely many zeros on the critical line [4]; Levinson and Conrey give positive proportions on the line [6, 7]; Speiser gives an equivalent form involving ζ' [5]; and Davenport–Heilbronn show that the functional equation and an Euler-product-type structure alone do not suffice to force zeros onto the line: Dirichlet series of the same type can have off-line zeros [8].

Lean formalization. The items above are covered by the files `ValueBridge.lean`, `TCoordOffline.lean`, `FoldObservation.lean` and `proof.lean`; each file is at 0 `sorry`.

Conclusion. The remaining step is not the gap of a single lemma but a universal constraint equivalent to the Riemann hypothesis; the rigid projection that links an off-line zero to a contradiction is called a generator in this paper. This paper does not make the construction of the generator a project; it only records the existing structural facts and the boundary clearly.

9 Verification record and reproduction

Phenomenon. Every declaration cited in this paper compiles under one tool chain, and can be checked independently.

Record. Environment: Lean 4.32.2 and mathlib (revision 905b95818e) [13, 14]. All 0 error, 0 `sorry`; no axioms beyond those of mathlib are introduced; the files cited here do not depend on the only explicit axiom of the companion work (the Pat0 absorbing rule, located in the phase-locking chain). The declaration count counts theorems + lemmas + defs, not instances:

File	Declarations
`AxisExactObservation.lean`	18
`FoldObservation.lean`	90
`NativeAxisEquiv.lean`	11
`EulerFoldImagAxis.lean`	89
`ValueBridge.lean`	25
`PreciseMigration.lean`	20
`CommonLineCheck.lean` (probe)	2

Table 3: Files and declaration counts.

The count of the assembly chain in the companion tree: fourteen layers, 284 declarations in total (with cross-layer repeats; the main proof file `proof.lean` holds 231 under the same convention), 0 `sorry` (Figure 4). The last row of Table 3 is taken from the probe directory (an independently placed check item); the other files belong to the paper's companion tree. Reproduction: inside that tree run `lake env lean ../探针_20261010/CommonLineCheck.lean` (expected exit 0); library-level checks use `build.sh` and `check_sorry.py`.

Hypotheses. Version and caches are fixed as recorded; the numbers in Table 3 agree with the declaration counts of the files (convention as above).

Lean formalization. See Table 3; all 0 `sorry`.

Conclusion. The chain of evidence can be checked independently.

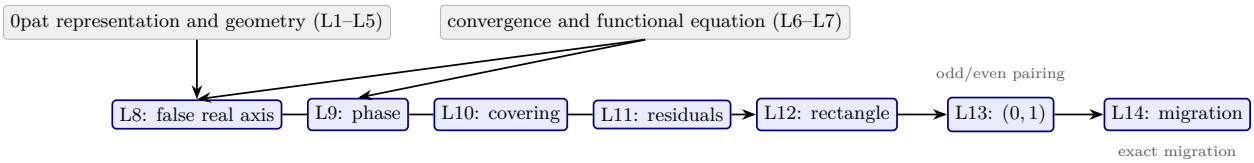


Figure 4: The fourteen layers of the assembly chain and the dependency flow (overview of the assembly chain in the companion tree); the declarations cited in this paper about the observation plane lie in L8 (false real axis) and in the migration layer L14.

10 Conclusion

Phenomenon. The paper has two parts. One part consists of verified equivalences and structures: the three coordinate properties of the observation plane, judgement completeness, and the reading form of the Riemann hypothesis. The other part is a universal constraint that remains open: the projection reading of every zero in the strip is zero.

Theorem (summary). The Riemann hypothesis is equivalent to "the projection reading of every zero in the strip is zero" (`fold_rh_iff_axis_form`); judgement completeness is `projection_zero_line_complete`; the reduction boundary writes "no off-line zeros", "all readings vanish" and "counting agrees" as one and the same thing.

Hypotheses (boundary). This paper does not claim to prove or disprove the Riemann hypothesis, nor does it make metamathematical assertions about provability; classical results that are not machine-verified are marked KNOWN; the coordinate properties and judgement completeness (§4 to §6) apply to all points or to individual zeros, and are statements of a different strength from the universal constraint (§7); they must not be conflated.

Lean formalization. All 0 error, 0 `sorry`; files and counts in §9.

Conclusion. The Riemann hypothesis can be placed precisely on a machine-verified observation plane: it is equivalent to "all readings of zeros are zero". The judgement layer is complete; the statement itself remains open. Further progress requires constructing a generator on the plane (see §8); everything before this step has been organized in this paper.

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